

GIRISH VUYYURU

Python AI Developer | Agentic AI Engineer | Multi-Agent Systems | LangGraph | LangChain | MCP | FastAPI

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gireeshvuyyuru501-design.github.io/Launch-AI-engineering-portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with 5 years of combined professional and advanced academic experience, with hands-on expertise in Python, Generative AI, Agentic AI, RAG, LangChain, LangGraph, MCP, FastAPI, cloud platforms, and production AI systems.

TECHNICAL SKILLS

Python & Backend: Python, FastAPI, Flask, Django, Pydantic, REST APIs, Microservices, OpenAPI/Swagger, Async Processing, OOP, SQL, Kafka, Redis, Streaming, Performance Optimization

Agentic AI & LLMs: Agentic AI, Generative AI, Multi-Agent Systems, LangChain, LangGraph, LangSmith, CrewAI, MCP, FastMCP, Agent Orchestration, Planning, Reasoning, Memory, Tool Calling, Function Calling, Human-in-the-Loop, ReAct-style Workflows, Prompt Engineering

RAG & Evaluation: RAG, GraphRAG, Embeddings, Semantic/Vector Search, Pinecone, ChromaDB, Milvus, FAISS, pgvector, Weaviate, Azure AI Search, Hybrid Search, Reranking, RAGAS, Groundedness, Hallucination Detection, AI Observability, Guardrails

Cloud & DevOps: AWS (Bedrock, SageMaker, Lambda, ECS, EKS, S3, CloudWatch, IAM), Azure (Azure OpenAI, Azure AI Search, Databricks, AKS, Azure DevOps), GCP (Vertex AI, BigQuery, Cloud Run, GKE, Gemini), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, MLflow, OpenTelemetry, Grafana, Prometheus

Data & ML: Snowflake, Snowflake Cortex, Microsoft Fabric, Fabric Lakehouse, OneLake, PySpark, Airflow, Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers

PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer

Mastercard | Jan 2025 – Present | Purchase, NY

- Architected and developed production-grade Agentic AI applications using Python, LangChain and LangGraph, implementing planning, reasoning, memory, tool/function calling, state management and autonomous multi-step workflows for enterprise financial services.
- Designed multi-agent systems with specialized agents, conditional routing, shared state, tool orchestration and human-in-the-loop controls using MCP and FastMCP standardized tool interfaces.
- Built enterprise RAG and GraphRAG pipelines integrating document ingestion, embeddings, semantic/vector search, knowledge graphs, metadata filtering, reranking and grounded LLM response generation.
- Developed scalable Python/FastAPI REST APIs and microservices integrating LLMs, AI agents, vector databases, Snowflake/Snowflake Cortex and enterprise data sources; implemented SSE streaming for real-time GenAI experiences.
- Implemented LLM evaluation measuring groundedness, relevance, hallucination risk, tool-call accuracy, latency and response quality using RAGAS and LangSmith; added prompt-injection defenses, PII redaction, access controls and audit logging.
- Built production observability with OpenTelemetry, Grafana and Prometheus; containerized services with Docker/Kubernetes and automated deployments through Azure DevOps CI/CD.
- Owned AI features across requirements, architecture, development, testing, debugging, deployment, monitoring and production support while collaborating with architects, product owners, data scientists and platform teams.

Environment: Python, LangChain, LangGraph, LangSmith, MCP, FastMCP, RAG, GraphRAG, Snowflake Cortex, Microsoft Fabric, Azure OpenAI, Claude, Gemini, RAGAS, FastAPI, OpenTelemetry, Grafana, Prometheus, Docker, Kubernetes, Azure DevOps, MLflow

ML Engineer / AI Engineer

Veterans Affairs Department | Jan 2023 – Dec 2024 | Washington, D.C.

- Built Python-based Agentic AI and RAG applications using LangChain and OpenAI APIs, integrating retrieval, prompt workflows and enterprise data access for production AI use cases.
- Designed end-to-end AI/ML pipelines covering ingestion, preprocessing, feature engineering, model training, evaluation and production deployment on AWS SageMaker.

- Implemented cloud-native AI services on AWS Lambda, ECS and EKS; developed FastAPI integrations and applied IAM, OAuth and least-privilege security patterns for AI APIs.
- Built production evaluation and observability using CloudWatch and Prometheus and developed distributed data pipelines with AWS Glue, PySpark and Apache Airflow.
- Containerized services with Docker and automated build, testing and deployment workflows using Jenkins and GitHub Actions.

Environment: Python, LangChain, RAG, OpenAI, AWS, SageMaker, Lambda, ECS, EKS, Glue, PySpark, Airflow, FastAPI, Docker, Jenkins, GitHub Actions, CloudWatch, Prometheus, MLflow, IAM, OAuth

Data Scientist

Radixweb | Jan 2021 – Dec 2021 | India

- Developed Python-based machine learning solutions using PyTorch, TensorFlow, Keras, Scikit-learn and XGBoost and deployed production inference services on AWS SageMaker.
- Built Python/FastAPI REST APIs with Pydantic and OpenAPI/Swagger, processed large datasets using PySpark and implemented ML lifecycle management with MLflow.
- Implemented Docker containerization and GitHub Actions CI/CD for repeatable testing and deployment of AI/ML services.

Environment: Python, PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, FastAPI, Pydantic, REST APIs, AWS SageMaker, MLflow, PySpark, SQL, Docker, GitHub Actions

EDUCATION

Master of Science, Computer Science - Texas A&M University-Kingsville | Jan 2022 - Dec 2022 | GPA: 4.0/4.0

Bachelor of Technology, Computer Science & Engineering - Vignan's University, Guntur, Andhra Pradesh

CERTIFICATIONS

Anthropic (16 Certifications): MCP Introduction, MCP Advanced Topics, Claude Code, Agent Skills & Subagents, Building Effective Agents, Prompt Engineering, RAG & Retrieval Systems, LLM Evaluation, AI Safety Fundamentals, Responsible AI, Claude API Fundamentals, Tool Use & Function Calling, Constitutional AI, Multi-Agent Systems, LangChain Integration, Production AI Deployment